## VELS INSTITUTE OF SCIENCE, TECHNOLOGY AND ADVANCED STUDIES

**DEPARTMENT: Computer Science and Information Technology**
**SEMESTER: II**
**TITLE OF THE COURSE: Cloud Computing**

**COURSE CODE:24DMIT21**

**TIME: 03 Hours** **Max. Marks: 100**

### SECTION – A (3 MARKS)

| S. No | Questions | Unit | K-Level | CO | PO |
|---|---|---|---|---|---|
| 1. | Define cloud computing. | I | K1 | CO1 | PO1 |
| 2. | What are the key characteristics of cloud computing? | I | K1 | CO1 | PO1 |
| 3. | What is the NIST Cloud Computing Reference Architecture? | I | K1 | CO1 | PO1 |
| 4. | List different cloud service models. | I | K1 | CO1 | PO1 |
| 5. | Differentiate between public and private cloud. | I | K2 | CO2 | PO2 |
| 6. | What is the cloud ecosystem? | I | K1 | CO1 | PO1,PO6 |
| 7. | Define computing on demand. | I | K1 | CO1 | PO1 |
| 8. | What are system models for distributed computing? | I | K1 | CO1 | PO1 |
| 9. | What is the importance of cloud service management? | I | K1 | CO1 | PO1,PO4 |
| 10. | What are the benefits of using cloud computing? | I | K2 | CO2 | PO2 |
| 11. | Define the term "Cloud Solutions." | I | K1 | CO1 | PO1 |
| 12. | What is IaaS in cloud computing? | I | K1 | CO1 | PO1 |
| 13. | Define PaaS and give an example. | I | K2 | CO2 | PO2 |
| 14. | What is SaaS? Mention its advantages. | I | K2 | CO2 | PO2 |
| 15. | List major cloud service providers. | I | K1 | CO1 | PO1 |
| 16. | What are the different cloud deployment models? | I | K2 | CO2 | PO1,PO2, PO3 |
| 17. | What is the concept of virtualization in cloud computing? | I | K2 | CO2 | PO2 |
| 18. | What are the security concerns in cloud | I | K3 | C03 | PO3 |

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| | computing? | | | | |
| 19. | Define hybrid cloud with an example. | I | K2 | CO2 | PO2 |
| 20. | What are the challenges in cloud adoption? | I | K3 | CO3 | PO3 |
| 21. | What is virtualization? | II | K2 | CO3 | PO3 |
| 22. | What is the importance of virtualization in cloud computing? | II | K2 | CO2 | PO2 |
| 23. | List different types of virtualization. | II | K1 | CO3 | PO3 |
| 24. | Define hardware virtualization. | II | K1 | CO3 | PO3 |
| 25. | What is software virtualization? | II | K1 | CO3 | PO3 |
| 26. | Define server virtualization. | II | K1 | CO3 | PO3 |
| 27. | What is storage virtualization? | II | K1 | CO3 | PO3 |
| 28. | Define network virtualization. | II | K1 | CO3 | PO3 |
| 29. | What is desktop virtualization? | II | K1 | CO3 | PO3 |
| 30. | What are the benefits of virtualization? | II | K2 | CO3 | PO3 |
| 31. | Define virtual clusters. | II | K1 | CO3 | PO3 |
| 32. | Define virtualization of CPU. | II | K1 | CO3 | PO3 |
| 33. | What is memory virtualization? | II | K1 | CO3 | PO3 |
| 34. | Define virtualization of I/O devices. | II | K1 | CO3 | PO3 |
| 35. | Define resource management in virtualization. | II | K2 | CO3 | PO3 |
| 36. | What are virtualization tools and mechanisms? | II | K2 | CO3 | PO3 |
| 37. | Define implementation levels of virtualization. | II | K2 | CO3 | PO3 |
| 38. | What is virtual machine migration? | II | K2 | CO3 | PO3 |
| 39. | What are the challenges in virtualization? | II | K3 | CO3 | PO4 |
| 40. | Define data center automation using virtualization. | II | K2 | CO3 | PO4 |
| 41. | Define cloud infrastructure. | III | K1 | CO3 | PO3 |

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| 42. | What is the role of compute and storage clouds? | III | K1 | CO3 | PO4 |
| 43. | List the key components of cloud infrastructure. | III | K1 | CO3 | PO4 |
| 44. | What are the significance of resource provisioning in cloud computing? | III | K2 | CO4 | PO3 |
| 45. | What is inter-cloud resource management? | III | K2 | CO4 | PO3 |
| 46. | Mention two design challenges in cloud infrastructure. | III | K3 | CO4 | PO3 |
| 47. | Define layered cloud architecture. | III | K1 | CO4 | PO3 |
| 48. | What are the enabling technologies for the Internet of Things (IoT)? | III | K1 | CO4 | PO3 |
| 49. | How does cloud computing benefit IoT applications? | III | K2 | CO4 | PO3 |
| 50. | What is resource pooling in cloud computing? | III | K2 | CO4 | PO3 |
| 51. | Define platform deployment in cloud infrastructure. | III | K1 | CO4 | PO1 |
| 52. | Differentiate between public cloud and private cloud infrastructure. | III | K2 | CO4 | PO3 |
| 53. | What is  the concept of cloud resource sharing? | III | K2 | CO4 | PO3 |
| 54. | What is the importance of virtualization in cloud infrastructure? | III | K2 | CO4 | PO3 |
| 55. | How does scalability affect cloud infrastructure? | III | K3 | CO4 | PO2 |
| 56. | Mention two innovative applications of IoT. | III | K1 | CO4 | PO3 |
| 57. | Define cloud security challenges in IoT. | III | K3 | CO4 | PO3 |
| 58. | What is fog computing? | III | K1 | CO4 | PO3 |
| 59. | What is the role of edge computing in IoT? | III | K2 | CO4 | PO1 |
| 60. | List two advantages of cloud-based IoT solutions. | III | K1 | CO5 | PO4 |
| 61. | Define Map Reduce and its significance in cloud computing. | IV | K1 | CO4 | PO1 |
| 62. | What is Hadoop, and why is it used? | IV | K1 | CO4 | PO2 |
| 63. | What is the role of Apache Hadoop in cloud computing? | IV | K1 | CO4 | PO1 |
| 64. | What are the main components of the Map Reduce framework? | IV | K2 | CO5 | PO3 |

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| 65. | What is Twister, and how does it differ from Map Reduce? | IV | K2 | CO5 | PO4 |
| 66. | How does iterative Map Reduce improve performance? | IV | K3 | CO5 | PO4 |
| 67. | List two advantages of using Google App Engine for cloud applications. | IV | K1 | CO5 | PO4 |
| 68. | What is the role of Amazon AWS in cloud computing? | IV | K1 | CO5 | PO4 |
| 69. | Define cloud software environments with examples. | IV | K1 | CO5 | PO4 |
| 70. | What is OpenStack, and why is it used in cloud computing? | IV | K1 | CO5 | PO4 |
| 71. | Compare Eucalyptus and Open Nebula. | IV | K2 | CO5 | PO4 |
| 72. | How does Cloud Sim help in cloud computing research? | IV | K2 | CO5 | PO4 |
| 73. | What are the primary programming support tools for cloud computing? | IV | K2 | CO5 | PO4 |
| 74. | Define parallel programming paradigms in cloud computing. | IV | K1 | CO5 | PO3 |
| 75. | Define the term "mapping applications" in cloud computing. | IV | K1 | CO5 | PO4 |
| 76. | What are the advantages of using Aneka for cloud computing? | IV | K2 | CO5 | PO4 |
| 77. | How does virtualization contribute to cloud programming models? | IV | K2 | CO5 | PO4 |
| 78. | Define cluster computing and its relevance to cloud computing. | IV | K1 | CO5 | PO4 |
| 79. | What are the challenges faced in cloud programming models? | IV | K3 | CO5 | PO4 |
| 80. | How distributed programming enhances cloud performance? | IV | K3 | CO5 | PO4 |
| 81. | Define cloud security and its importance. | V | K1 | CO5 | PO3 |
| 82. | List common security challenges in cloud computing. | V | K2 | CO5 | PO4 |
| 83. | What is Software-as-a-Service (SaaS) security? | V | K1 | CO5 | PO4 |
| 84. | What is the importance of risk management in cloud security? | V | K2 | CO6 | PO4 |
| 85. | Define identity management in cloud security. | V | K2 | CO6 | PO4 |
| 86. | What are the key principles of cloud security governance? | V | K2 | CO6 | PO4 |
| 87. | Define the need for security monitoring in cloud computing. | V | K2 | CO6 | PO4 |
| 88. | What is autonomic security in cloud | V | | | |

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| | computing? | | K2 | CO6 | PO4 |
| 89. | Define virtual machine security. | V | K2 | CO6 | PO4 |
| 90. | What is the role of access control in cloud security? | V | K2 | CO6 | PO4 |
| 91. | What is data security in cloud computing. | V | K2 | CO6 | PO4 |
| 92. | How does encryption enhance cloud security? | V | K3 | CO6 | PO4 |
| 93. | List any two common cloud security threats. | V | K1 | CO6 | PO4 |
| 94. | What is the difference between private and public cloud security? | V | K2 | CO6 | PO4 |
| 95. | How does authentication work in cloud security? | V | K2 | CO6 | PO4 |
| 96. | Define security architecture design in the cloud. | V | K2 | CO6 | PO4 |
| 97. | What are the advantages of cloud security frameworks? | V | K3 | CO6 | PO4 |
| 98. | What is the concept of application security in cloud environments? | V | K2 | CO6 | PO4 |
| 99. | What is the importance of compliance in cloud security? | V | K2 | CO6 | PO4 |
| 100. | How does cloud security affect business operations? | V | K3 | CO6 | PO4 |

# SECTION – B (8 MARKS)

| S.No | Questions | Unit | K Level | CO | PO |
|---|---|---|---|---|---|
| 101. | Explain in detail the characteristics of cloud computing. | I | K2 | CO1 | PO1 |
| 102. | Compare different cloud service models: IaaS, PaaS, and SaaS. | I | K2 | CO1 | PO2 |
| 103. | Explain the NIST cloud computing reference model with a neat diagram. | I | K2 | CO2 | PO2 |
| 104. | Discuss different cloud deployment models. | I | K2 | CO2 | PO3 |
| 105. | Describe the advantages and disadvantages of cloud computing. | I | K2 | CO1 | PO4 |
| 106. | Explain the system models for distributed and cloud computing. | I | K2 | CO2 | PO1 |

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| 107. | Discuss the concept of computing on demand. | I | K2 | CO1 | PO2 |
| 108. | Explain cloud ecosystem and its components. | I | K2 | CO2 | PO3 |
| 109. | Discuss the challenges in implementing cloud solutions. | I | K3 | CO5 | PO6 |
| 110. | Explain cloud service management in detail. | I | K3 | CO6 | PO5 |
| 111. | Compare public, private, and hybrid cloud with examples. | I | K2 | CO2 | PO2 |
| 112. | Explain the significance of virtualization in cloud computing. | I | K2 | CO3 | PO3 |
| 113. | Discuss the security aspects of cloud computing. | I | K3 | CO5 | PO5 |
| 114. | Describe cloud computing architecture in detail. | I | K3 | CO2 | PO3 |
| 115. | Explain how cloud computing enables business scalability. | I | K2 | CO1 | PO4 |
| 116. | Compare traditional computing with cloud computing. | I | K2 | CO1 | PO2 |
| 117. | Explain the different types of virtualization with examples. | II | K2 | CO3 | PO3 |
| 118. | Discuss the role of virtualization in cloud computing. | II | K2 | CO3 | PO4 |
| 119. | Explain the concept of hardware and software virtualization. | II | K2 | CO3 | PO1 |
| 120. | Describe server virtualization and its advantages. | II | K2 | CO3 | PO4 |
| 121. | Explain in detail the concept of storage virtualization. | II | K2 | CO3 | PO2 |
| 122. | Discuss the importance of network virtualization. | II | K2 | CO3 | PO4 |
| 123. | Explain virtual clusters and resource management. | II | K3 | CO3 | PO5 |
| 124. | Discuss virtualization of CPU and memory. | II | K3 | CO3 | PO5 |
| 125. | Explain I/O device virtualization. | II | K3 | CO3 | PO6 |
| 126. | Discuss the implementation levels of virtualization. | II | K3 | CO3 | PO5 |
| 127. | Explain various tools and mechanisms used in virtualization. | II | K3 | CO3 | PO4 |
| 128. | Discuss the concept of virtual machine migration and its benefits. | II | K3 | CO3 | PO5 |
| 129. | Explain the challenges associated with virtualization. | II | K4 | CO3 | PO6 |
| 130. | Discuss how virtualization enables data center | II | | | |

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| | automation. | | K3 | CO3 | PO4 |
| 131. | Compare different virtualization technologies. | II | K4 | CO3 | PO2 |
| 132. | Explain how virtualization improves IT infrastructure management. | II | K3 | CO3 | PO5 |
| 133. | Explain the architectural design of compute and storage clouds. | III | K2 | CO2 | PO1 |
| 134. | Describe the layered cloud architecture development. | III | K2 | CO2 | PO2 |
| 135. | What are the major design challenges in cloud infrastructure? | III | K4 | CO6 | PO3 |
| 136. | Explain inter-cloud resource management and its significance. | III | K3 | CO2 | PO4 |
| 137. | Discuss the process of resource provisioning in cloud computing. | III | K3 | CO2 | PO5 |
| 138. | Explain the importance of platform deployment in cloud infrastructure. | III | K2 | CO2 | PO4 |
| 139. | How do cloud resources enable IoT technologies? | III | K2 | CO2 | PO6 |
| 140. | Discuss the role of virtualization in cloud infrastructure and IoT. | III | K3 | CO3 | PO5 |
| 141. | Explain the security challenges in cloud-based IoT applications. | III | K4 | CO5 | PO6 |
| 142. | Differentiate between fog computing and edge computing. | III | K3 | CO2 | PO2 |
| 143. | How does cloud computing improve resource efficiency in IoT? | III | K2 | CO2 | PO4 |
| 144. | Explain the process of global exchange of cloud resources. | III | K3 | CO2 | PO6 |
| 145. | Describe cloud ecosystem and its relevance in IoT. | III | K2 | CO2 | PO5 |
| 146. | What are the key advantages of using cloud computing for IoT? | III | K2 | CO2 | PO4 |
| 147. | Discuss service management in cloud infrastructure. | III | K3 | CO6 | PO3 |
| 148. | Explain the impact of cloud computing on smart cities. | III | K3 | CO2 | PO5 |
| 149. | Explain in detail the working of the MapReduce model with an example. | IV | K3 | CO4 | PO1 |
| 150. | Describe the Hadoop architecture and its key components. | IV | K3 | CO4 | PO5 |
| 151. | Compare and contrast Twister and Iterative MapReduce. | IV | K4 | CO4 | PO2 |
| 152. | Explain the role of Google App Engine in cloud computing. | IV | K2 | CO4 | PO3 |
| 153. | Discuss different programming paradigms used in | IV | | | |

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| | cloud computing. | | K3 | CO4 | PO4 |
| 154. | Compare OpenStack and CloudSim with their advantages. | IV | K4 | CO4 | PO2 |
| 155. | Explain the features of Eucalyptus and Open Nebula. | IV | K3 | CO4 | PO5 |
| 156. | Discuss the importance of software environments in cloud computing. | IV | K2 | CO4 | PO6 |
| 157. | Explain how Amazon AWS supports cloud-based applications. | IV | K3 | CO4 | PO5 |
| 158. | Describe various tools available for cloud programming support. | IV | K2 | CO4 | PO4 |
| 159. | How does parallel programming improve cloud performance? | IV | K4 | CO4 | PO5 |
| 160. | Explain the concept of data locality in MapReduce. | IV | K3 | CO4 | PO2 |
| 161. | Compare different types of virtualization in cloud computing. | IV | K3 | CO3 | PO4 |
| 162. | Discuss the impact of cloud software environments on application development. | IV | K4 | CO4 | PO3 |
| 163. | Discuss different programming paradigms used in cloud computing. | IV | K3 | CO4 | PO4 |
| 164. | Compare OpenStack and CloudSim with their advantages. | IV | K4 | CO4 | PO2 |
| 165. | Explain the security risks associated with cloud computing. | V | K3 | CO5 | PO1 |
| 166. | Discuss various cloud security challenges and their mitigation techniques. | V | K4 | CO5 | PO6 |
| 167. | Explain the significance of identity and access management in cloud security. | V | K3 | CO5 | PO2 |
| 168. | How does risk management improve cloud security? | V | K5 | CO5 | PO6 |
| 169. | Describe different types of cloud security threats with examples. | V | K3 | CO5 | PO5 |
| 170. | Explain the role of encryption in protecting cloud data. | V | K3 | CO5 | PO4 |
| 171. | Compare security challenges in SaaS, PaaS, and IaaS. | V | K4 | CO5 | PO2 |
| 172. | Discuss best practices for securing virtual machines in the cloud. | V | K3 | CO7 | PO5 |
| 173. | Explain the concept of autonomic security and its applications. | V | K3 | CO7 | PO4 |
| 174. | How does security monitoring help in preventing cyber threats in the cloud? | V | K3 | CO7 | PO6 |
| 175. | Describe cloud security governance and its importance. | V | K3 | CO7 | PO5 |
| 176. | Discuss various security mechanisms used in | V | K3 | CO7 | PO6 |

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| | cloud computing. | | | | |
| 177. | How do firewalls and intrusion detection systems protect cloud data? | V | K3 | CO7 | PO4 |
| 178. | Explain security architecture design in cloud computing. | V | K4 | CO7 | PO3 |
| 179. | Describe how compliance requirements impact cloud security strategies. | V | K3 | CO5 | PO2 |
| 180. | Explain different techniques used for cloud application security. | V | K3 | CO7 | PO5 |

# SECTION – C (15 MARKS)

| S. No | Questions | Unit | K Level | CO | PO |
|---|---|---|---|---|---|
| 181. | Explain cloud computing architecture with a neat diagram. | I | K2 | CO2 | PO1 |
| 182. | Discuss different cloud models and compare their features. | I | K2 | CO2 | PO1 |
| 183. | Explain in detail the concept of virtualization and its impact on cloud computing. | I | K2 | CO3 | PO1 |
| 184. | Describe in detail the security challenges in cloud computing and how to overcome them. | I | K3 | CO5 | PO1 |
| 185. | Discuss the NIST cloud computing reference architecture in depth. | I | K2 | CO2 | PO1 |
| 186. | Explain the role of cloud computing in modern IT infrastructure. | I | K2 | CO2 | PO1 |
| 187. | Discuss in detail the cloud service models (IaaS, PaaS, SaaS) with real-world examples. | I | K3 | CO2 | PO1 |
| 188. | Explain how cloud computing is transforming industries and businesses. | I | K3 | CO2 | PO1 |
| 189. | Explain in detail the concept, types, and implementation of virtualization. | II | K3 | CO3 | PO1 |
| 190. | Discuss the impact of virtualization on cloud computing. | II | K3 | CO3 | PO1 |
| 191. | Explain server, storage, and network virtualization in depth. | II | K3 | CO3 | PO1 |
| 192. | Discuss the role of virtual clusters in cloud computing. | II | K3 | CO3 | PO1 |
| 193. | Explain the process of virtualization of CPU, memory, and I/O devices. | II | K3 | CO3 | PO1 |
| 194. | Describe the implementation levels of virtualization and its benefits. | II | K3 | CO3 | PO1 |
| 195. | Explain the challenges and security concerns in virtualization. | II | K3 | CO3 | PO5 |
| 196. | Discuss the role of virtualization in data center automation. | II | K2 | CO3 | PO4 |

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| 197. | Explain in detail the layered cloud architecture with examples. | III | K3 | CO2 | PO1 |
| 198. | Discuss the challenges and solutions in designing cloud infrastructure. | III | K4 | CO2 | PO3 |
| 199. | How does inter-cloud resource management enhance cloud computing performance? | III | K4 | CO2 | PO3 |
| 200. | Explain the concept of cloud-based IoT and its innovative applications. | III | K3 | CO2 | PO1 |
| 201. | Discuss the role of virtualization in cloud infrastructure and its impact on IoT. | III | K3 | CO2 | PO1 |
| 202. | Describe various cloud resource management techniques with suitable examples. | III | K3 | CO2 | PO3 |
| 203. | Explain in detail the importance of security and privacy in cloud-based IoT applications. | III | K4 | CO5 | PO5 |
| 204. | Discuss the architectural design and deployment models of cloud infrastructure in IoT. | III | K3 | CO2 | PO4 |
| 205. | Explain in detail the Hadoop framework and its role in cloud computing. | IV | K3 | CO4 | PO4 |
| 206. | Compare and contrast different cloud software environments like Eucalyptus, OpenStack, and Open Nebula. | IV | K4 | CO4 | PO4 |
| 207. | Discuss in detail how MapReduce is used for big data processing. | IV | K3 | CO4 | PO4 |
| 208. | Explain the significance of virtualization in cloud computing with examples. | IV | K3 | CO3 | PO1 |
| 209. | Describe cloud programming models and their applications. | IV | K3 | CO4 | PO4 |
| 210. | Discuss different parallel programming paradigms in the context of cloud computing. | IV | K3 | CO4 | PO4 |
| 211. | Explain the role of Amazon AWS and Google App Engine in cloud-based solutions. | IV | K3 | CO4 | PO4 |
| 212. | Analyze the challenges and future trends in cloud programming models. | IV | K4 | CO4 | PO5 |
| 213. | Discuss in detail the major security challenges in cloud computing and their solutions. | V | K4 | CO5 | PO5 |
| 214. | Explain cloud security governance and the role of risk management in detail. | V | K5 | CO7 | PO6 |
| 215. | Analyze the importance of encryption and authentication in cloud security. | V | K5 | CO7 | PO5 |
| 216. | Explain the different layers of cloud security and their impact. | V | K3 | CO7 | PO6 |
| 217. | Discuss access control mechanisms used in cloud security with examples. | V | K3 | CO7 | PO6 |
| 218. | Compare and contrast various cloud security architectures. | V | K4 | CO7 | PO6 |
| 219. | Explain the role of security monitoring and incident response in cloud security. | V | K3 | CO7 | PO5 |

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| 220. | Describe best practices and strategies for securing cloud infrastructure. | V | K5 | CO7 | PO6 |

**K1- Remember; K2- Understand; K3- Apply; K4- Analyze;**

**K5-Evaluate; K6-Create**